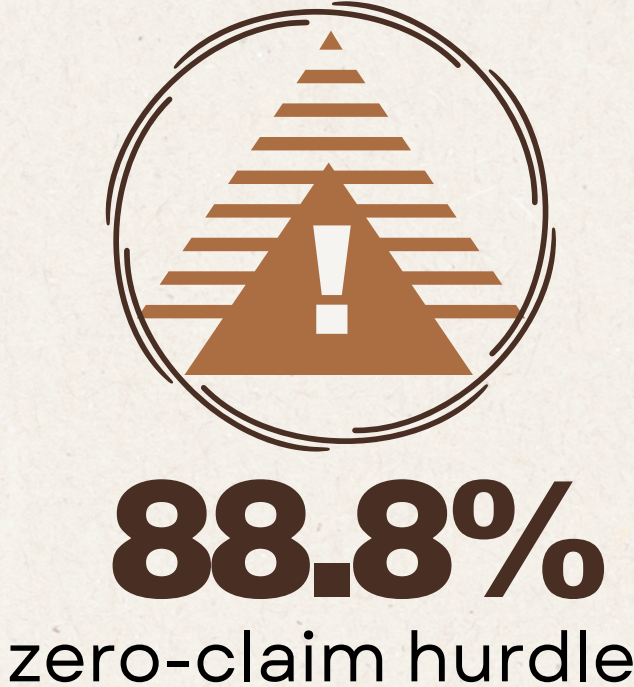


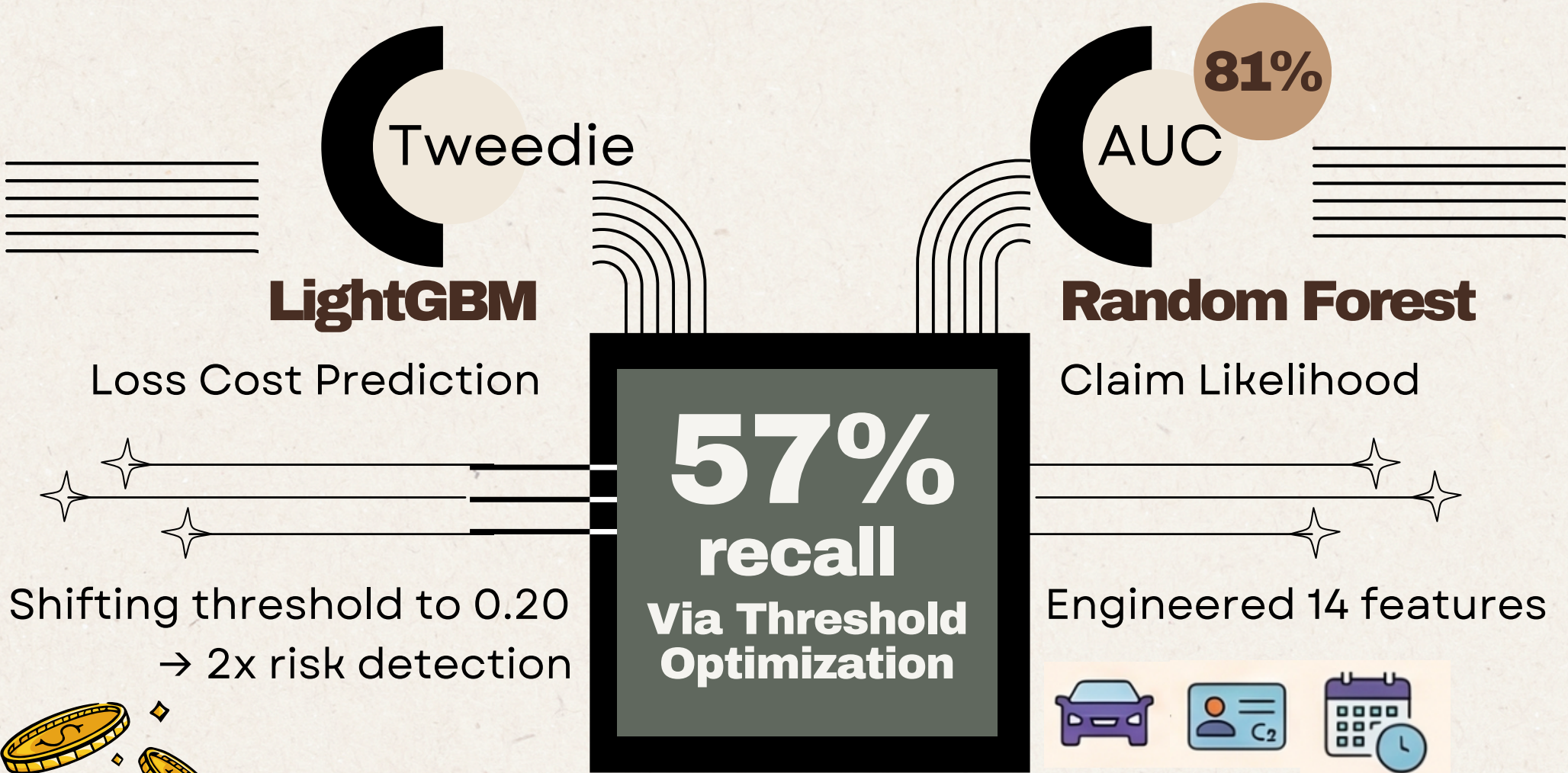
Optimizing Insurance Risk via Machine Learning

The Challenge



Overcharging low-risk clients drives them away -> portfolio unprofitable

The Solution



The Insights



regression

performance

	LC MSE	HALC MSE
Tweedie GLM (Baseline)	262,958	1,098,788
🏆 LightGBM (Winner)	259,627	1,026,332
Random Forest	261,284	1,029,447

risk-aligned pricing

Net Premium aligns with predicted loss → current premium structure
Vehicle weight & power-to-weight ratio → mechanical risk